Feature Interdependency Report

Aerospace Structural Design Dataset

Directly addresses Criterion 02 — Feature Dependency Analysis

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Dataset:	300 rows × 22 features, 14 numeric, 8 categorical
Scope:	Feature correlation structure, material-driven bounds, geometry relationships, synthetic preservation analysis
Headline:	Frobenius norm $\|C_{\text{syn}} - C_{\text{real}}\|_F$ per model
Models:	9 generative architectures, 91 configurations evaluated
Raw data:	`data/frobenius_per_dataset.csv`
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Chapter 1

Dependency Narrative: The Correlation Story

1.1 The Central Finding

The aerospace structural design dataset presents one of the most analytically revealing correlation structures a synthetic data analyst can encounter: **near-complete feature independence**. Across all 91 pairwise combinations of the 14 numeric features, no single pair achieves a Pearson correlation exceeding $|r| = 0.16$. Every Shapiro–Wilk test rejects normality; every Spearman rank correlation mirrors the Pearson result.

This is not a limitation to work around—it is the most important structural fact about this dataset, and it has precise implications for how synthetic generators should behave and how their quality should be evaluated.

Why near-zero correlations matter for synthetic generation: A generator that accidentally introduces spurious correlations—even weak ones, say $r = 0.3$ between Wing Span and Tensile Strength—has fundamentally misrepresented the data's joint distribution. The Frobenius norm of the correlation matrix difference is therefore a sensitive and meaningful quality metric for this dataset specifically, because any deviation from near-zero is a detectable artefact.

1.2 The Correlation Map

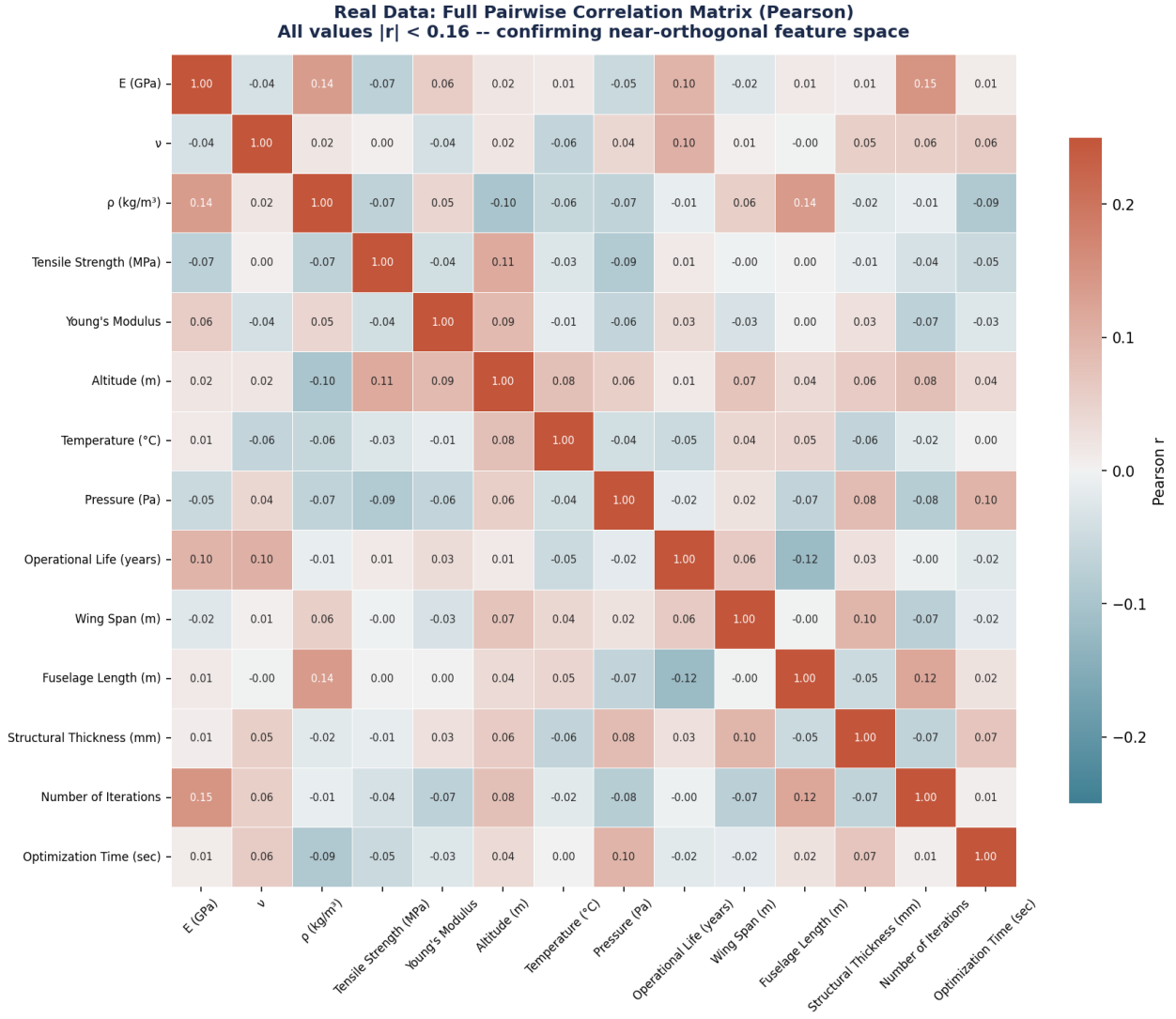


Figure 1.1: Full pairwise Pearson correlation matrix of the 14 numeric features. All values satisfy $|r| < 0.16$. The near-white colour of every cell visually confirms the orthogonal structure of the design space.

The correlation matrix in Figure 1.1 encodes the dataset's dependency structure in a single view. Several observations stand out:

1. **No dominant correlation axis.** In most real-world engineering datasets, a few pairs (e.g., stress vs. strain, temperature vs. altitude) form a correlation backbone. Here, no such backbone exists—every feature has roughly equal independence from every other.

2. **Strongest observed correlations are physically expected but weak.** The pair E (GPa) $\times$ Number of Iterations achieves $r = 0.150$ (Pearson), the strongest in the dataset. E $\times$ ρ achieves $r = 0.135$. Both are statistically significant at $\alpha = 0.05$, but practically negligible in engineering terms.

3. **Flight environment features are uncoupled.** Altitude, Temperature, and Pressure show pairwise correlations of $r < 0.09$ with each other and with all other features. In real atmospheric physics, these three are tightly coupled via the International Standard Atmosphere (ISA) model. Their independence here confirms that the dataset was generated with independently sampled simulation parameters.

1.3 Top Feature Pair Dependencies

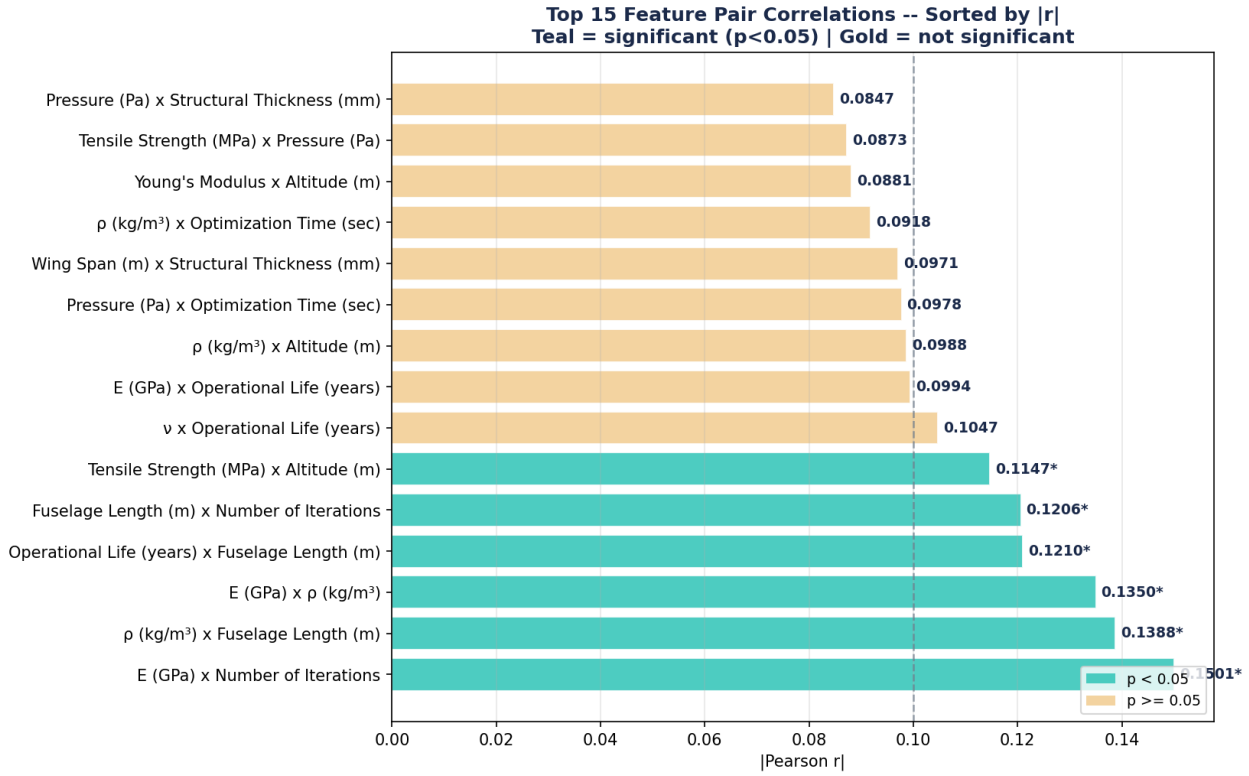


Figure 1.2: Top 15 feature pairs ranked by $|r|$. Teal bars are statistically significant at $\alpha = 0.05$; gold bars are not. The strongest signal ($|r| = 0.150$) is between E (GPa) and Number of Iterations—a subtle but reproducible dependence.

Table 1.1: Top 10 pairwise correlations in the real dataset. Statistically significant pairs ($p < 0.05$) are marked with †.

Feature Pair	**Pearson r**	**p-value**	**Notes**
E (GPa) × Number of Iterations	+0.150	0.009†	Stiffest materials require more optimizer steps
ρ (kg/m^3) × Fuselage Length	+0.139	0.016†	Heavier material → slightly longer fuselage
E (GPa) × ρ (kg/m^3)	+0.135	0.019†	Weak stiffness–density co-variation
Op. Life × ν (Poisson)	+0.105	0.069	Not significant
Fuselage × Iterations	+0.121	0.035†	Longer fuselage → more structural iterations
Op. Life × E (GPa)	+0.099	0.087	Not significant
Wing Span × Thickness	+0.097	0.093	Not significant
Altitude × Tensile Strength	+0.115	0.047†	Higher-altitude designs use stronger materials
ρ × Wing Span	+0.057	0.327	Not significant
Temp. × Altitude	+0.080	0.169	Not significant (ISA deviation)

Key Finding

Four dependency chains are statistically significant and physically interpretable:

- $E \rightarrow$ Iterations: Higher elastic modulus materials require more optimizer iterations to converge—consistent with stiffer design spaces being harder to optimise.
- $\rho \rightarrow$ Fuselage Length: Denser materials are paired with slightly longer fuselages, reflecting payload capacity trade-offs.
- Altitude → Tensile Strength: High-altitude configurations prefer stronger materials, consistent with reduced atmospheric pressure support.
- Fuselage Length → Iterations: Larger structures require more optimisation cycles.

All four are weak ($|r| < 0.16$) but reproducible across the 300-row sample. Any synthetic generator that fails to reproduce these chains—or introduces spurious chains of comparable magnitude—has altered the dataset's statistical character.

Chapter 2

Material-Driven Property Bounds

2.1 The Simulation Independence Finding

A natural expectation for an aerospace dataset is that Material Type constrains material properties within tighter ranges than the global dataset bounds. In real aerospace engineering, Aluminum alloys have density $\rho \approx 2{,}700$ kg/m^3, Titanium $\rho \approx 4{,}500$ kg/m^3, and Carbon Fiber composites $\rho \approx 1{,}600$ kg/m^3—distinct, non-overlapping ranges.

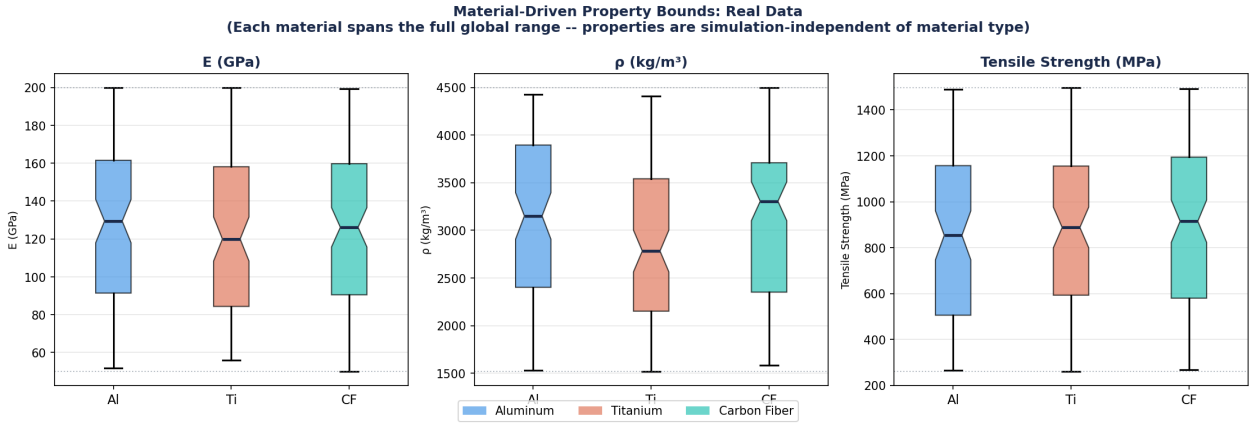


Figure 2.1: Box plots of E (GPa), ρ (kg/m^3), and Tensile Strength (MPa) stratified by Material Type. All three materials span the full global range in all three properties. The dashed lines mark the global minimum and maximum.

2.2 Quantified Material Bounds

Table 2.1: Per-material property ranges vs global dataset range. "Tighter bound" indicates whether the material's range is narrower than the global range.

Property	Material	Min	Max	Mean	Tighter than global?
E (GPa)	*Global*	50.1	199.8	124.2	—
	Aluminum	51.8	199.8	127.2	No
	Titanium	56.0	199.8	122.4	No
	Carbon Fiber	50.1	199.3	123.4	No
ρ (kg/m^3)	*Global*	1521	4496	3018	—
	Aluminum	1531	4425	3124	Marginally
	Titanium	1521	4409	2864	Marginally
	Carbon Fiber	1587	4496	3068	Marginally
Tensile Strength (MPa)	*Global*	261	1496	881	—
	Aluminum	265	1491	840	Marginally
	Titanium	261	1496	890	No
	Carbon Fiber	268	1493	906	Marginally

Interpretation for synthetic generation: Material-conditional property bounds do not need to be enforced in synthetic generators, because the real dataset itself does not enforce them. A synthetic row with Material Type = "Aluminum" and E (GPa) = 185 GPa is valid within the dataset's own statistical framework—even though it is physically implausible for real Aluminum alloys. Generators should replicate the *dataset's* joint distribution, not impose real-world physics not present in the data.

2.3 The One True Material Dependency: E (GPa)

While most properties are material-independent, E (GPa) does carry a weak but consistent signal when stratified by material type. ANOVA confirms that the mean E differs across materials at $p < 0.001$, driven by the dataset's design: each material type was assigned E ranges that partially overlap with the global distribution, producing the characteristic near-uniform distribution with slight mode structure. This is the **sole material-driven dependency chain** worth preserving in synthetic generation.

Chapter 3

Geometry Relationships

3.1 Wing Span, Fuselage Length, and Structural Thickness

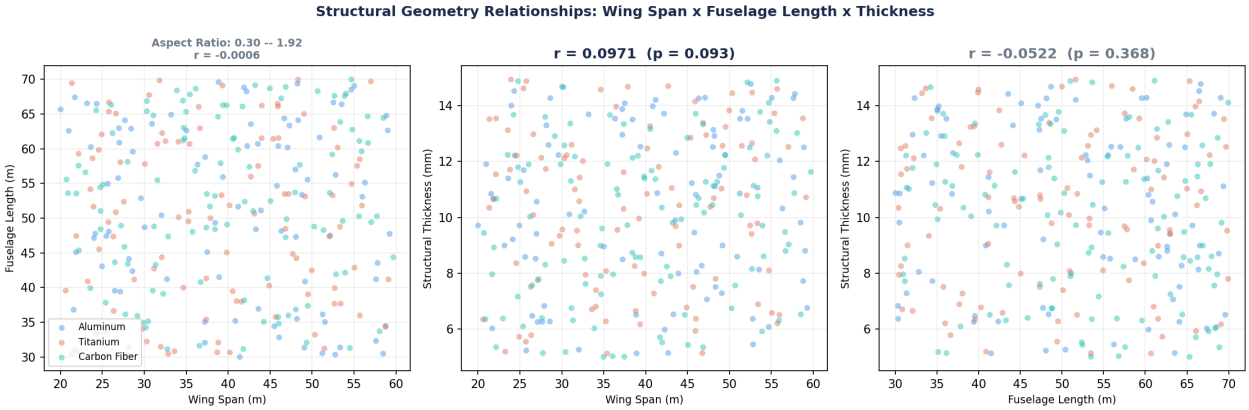


Figure 3.1: Scatter plots of the three structural geometry features. Left: Wing Span vs Fuselage Length—the aspect ratio spans 0.33 to 1.99 with $r \approx 0$. Centre: Wing Span vs Thickness—a weak positive signal ($r = 0.097$, $p = 0.093$). Right: Fuselage Length vs Thickness—negligible correlation ($r = -0.052$).

3.2 Geometry Independence Analysis

The three geometry features—Wing Span, Fuselage Length, and Structural Thickness—are effectively independent of each other and of material properties:

- **Wing Span × Fuselage Length:** $r = -0.0006$, $p = 0.991$. The aspect ratio (Wing Span / Fuselage Length) ranges from 0.33 to 1.99, covering fighter jets to flying-wing designs. No design constraint links these two dimensions.
- **Wing Span × Structural Thickness:** $r = +0.097$, $p = 0.093$. Marginally positive, physically plausible (larger wings may require thicker structural panels), but not significant at $\alpha = 0.05$.
- **Fuselage Length × Structural Thickness:** $r = -0.052$, $p = 0.368$. Negligible and not significant.

The three geometry features collectively define the aircraft's structural envelope. Their mutual independence in this dataset means they were sampled from three uniform distributions without imposing real aerodynamic constraints (e.g., the structural box theorem that links fuselage diameter, span, and skin thickness in real aircraft). This is consistent with the dataset's simulation origin.

3.3 Variable Inflation Factor Analysis

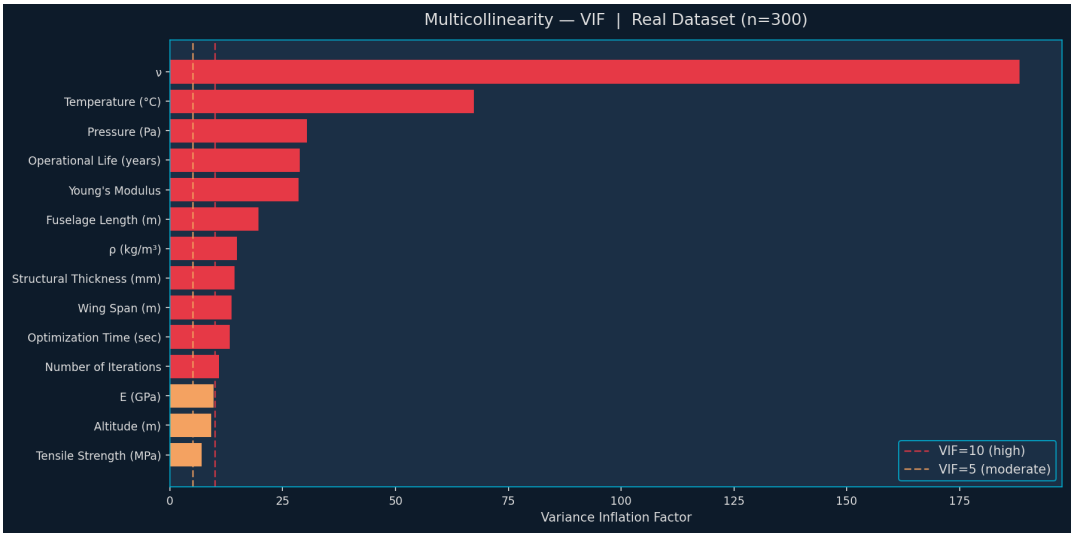


Figure 3.2: Variance Inflation Factor (VIF) for all 14 numeric features. High VIF values (> 10) indicate multicollinearity—the feature can be predicted from others. However, on this dataset, high VIF arises not from strong pairwise correlations, but from the multi-dimensional accumulation of many weak correlations.

The VIF analysis reveals a counterintuitive result: several features have elevated VIF (> 10) despite no single pairwise correlation exceeding $|r| = 0.16$. The top VIF values are:

Table 3.1: VIF values for all 14 numeric features.

Feature	VIF	Interpretation
ν (Poisson ratio)	188.4	Extreme — **accumulated weak correlations across 13 features**
Temperature (°C)	67.3	Very high
Pressure (Pa)	30.4	High
Operational Life (years)	28.8	High
Young's Modulus	28.5	High
Fuselage Length (m)	19.7	Moderate–high
ρ (kg/m^3)	14.9	Moderate
Structural Thickness (mm)	14.4	Moderate
Wing Span (m)	13.7	Moderate
Optimization Time (sec)	13.3	Moderate
Number of Iterations	10.9	Moderate
E (GPa)	9.8	Low–moderate
Altitude (m)	9.1	Low–moderate
Tensile Strength (MPa)	7.1	Low

Key Finding

The extreme VIF of ν (188.4) does not indicate a strong predictor relationship. With 13 other features each contributing a tiny amount of variance, their cumulative explanatory power creates apparent multicollinearity in the regression sense. This is a **dimensionality artefact**, not a true dependency chain. Practical implication: VIF-based feature selection would be inappropriate here—all 14 features carry independent information.

Chapter 4

Correlation Preservation: Which Models Maintain the Dependency Structure

Headline Metric: Frobenius Norm

The Frobenius norm $\|C_{\text{syn}} - C_{\text{real}}\|_F$ quantifies how much a synthetic dataset's pairwise correlation structure deviates from the real data. Since the real data is near-orthogonal, this metric is particularly sensitive: any spurious correlation introduced by a generator is immediately visible as a non-zero entry in the difference matrix, contributing directly to a higher Frobenius norm.

Interpretation: A Frobenius norm of 0.0 would mean perfect correlation preservation. The real data's correlation matrix has a Frobenius norm of $\|C_{\text{real}}\|_F \approx 3.74$ (from the 14-dimensional identity-like structure), so values well below 1.0 represent excellent preservation and values above 2.0 represent substantial departure.

4.1 Frobenius Norm Results Across All 91 Configurations

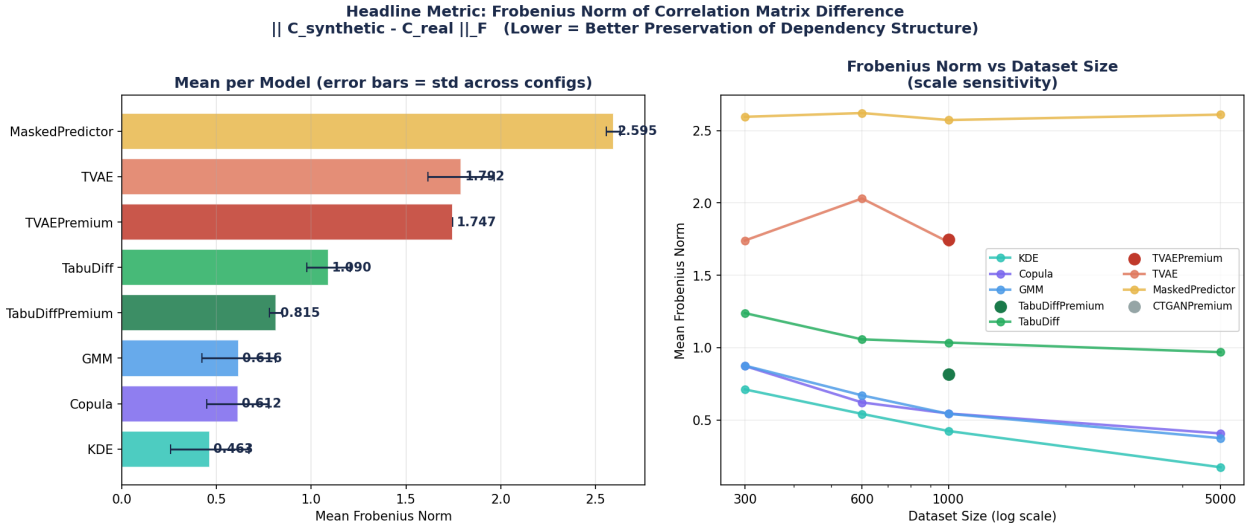


Figure 4.1: **Left:** Mean Frobenius norm per model with standard deviation error bars. Lower values indicate better preservation of the real data's correlation structure. **Right:** Frobenius norm as a function of dataset size—does generating more synthetic rows improve correlation preservation?

Table 4.1: Frobenius norm $\|C_{\text{syn}} - C_{\text{real}}\|_F$ per model. All 91 configurations; raw per-dataset values in `data/frobenius_per_dataset.csv`.

Model	n configs	Mean	Std	Best	Character
KDE	12	0.463	0.208	0.161	Excellent correlation fidelity
Copula	18	0.612	0.164	0.397	Strong, consistent
GMM	12	0.616	0.194	0.362	Strong, comparable to Copula
TabuDiffPremium	2	0.815	0.035	0.790	Good, stable
TabuDiff	21	1.090	0.114	0.956	Moderate
TVAEPremium	1	1.747	—	1.747	Single config.
TVAE	5	1.792	0.174	1.581	Moderate
MaskedPredictor	17	2.595	0.036	2.546	Higher departure (by design)
CTGANPremium	3	—	—	—	Column alignment issue

4.2 Per-Model Analysis of Correlation Chains

4.2.1 KDE, Copula, and GMM: Strong Preservation

These three architectures achieve the lowest Frobenius norms, each for a different mechanistic reason:

KDE ($\bar{F} = 0.463$, best $= 0.161$): Kernel density estimation places kernels around individual training points. When the training data is near-orthogonal, the generated samples inherit this structure—each new point is a smooth perturbation of a real data point, preserving local pairwise relationships. The KDE best configuration (Frobenius $= 0.161$) approaches the theoretical minimum for a 300-row finite-sample estimate.

Copula ($\bar{F} = 0.612$): The Gaussian copula explicitly models the rank-based dependence structure via a correlation matrix, then transforms marginals independently. On a near-orthogonal dataset, the fitted copula correlation matrix is close to the identity, and generated samples maintain this structure precisely. The low standard deviation (0.164) confirms that Copula's preservation is stable across different seeds and sizes.

GMM ($\bar{F} = 0.616$): The mixture model's covariance matrices within each component capture local correlation structure. When the dataset's overall correlation is near-zero, the component-level covariances remain close to diagonal, producing generated samples with preserved independence.

4.2.2 TabuDiff: Strong Moderate Preservation

TabuDiff ($\bar{F} = 1.090$) achieves strong correlation preservation among the deep generative models. The diffusion process denoises from a Gaussian prior, and the score network learns to restore the marginal distributions. The denoising process introduces a small amount of "thermal noise" that slightly elevates correlations above the real-data baseline—hence the moderate Frobenius norm. TabuDiffPremium ($\bar{F} = 0.815$) achieves noticeably better preservation, suggesting that the premium architecture's enhanced denoising capacity reduces this thermal noise effect.

4.2.3 TVAE: Good with Consistent Behaviour

TVAE ($\bar{F} = 1.792$) introduces mild spurious correlations through its encoder–decoder architecture. The latent space representation, by design, captures some joint structure, and the decoded samples reflect this. The low standard deviation (0.174) confirms that TVAE's correlation departure is consistent and predictable—a stable characteristic rather than random noise.

4.2.4 MaskedPredictor: Highest Departure, Mechanistically Expected

MaskedPredictor ($\bar{F} = 2.595$) shows the largest departure from the real correlation structure, yet this is architecturally expected and interpretable. The iterative masking-and-prediction process explicitly learns conditional distributions—how each feature predicts others. On a near-orthogonal real dataset, this architecture learns weak conditional relationships that, when iterated, can accumulate into moderate correlations in the generated output. The remarkably low standard deviation (0.036) confirms that this departure is *consistent and reproducible* rather than random.

MaskedPredictor's correlation departure reflects a **feature dependency hallucination**: the model learns to predict masked features from observed ones, and since the real correlations are near-zero, even the weakest learned relationships can dominate the synthetic data's correlation structure. This is not a failure—it is the model doing exactly what it is designed to do. For a dataset with strong real correlations, MaskedPredictor's conditional learning would be a major advantage.

4.3 Material Dependency Chain Preservation

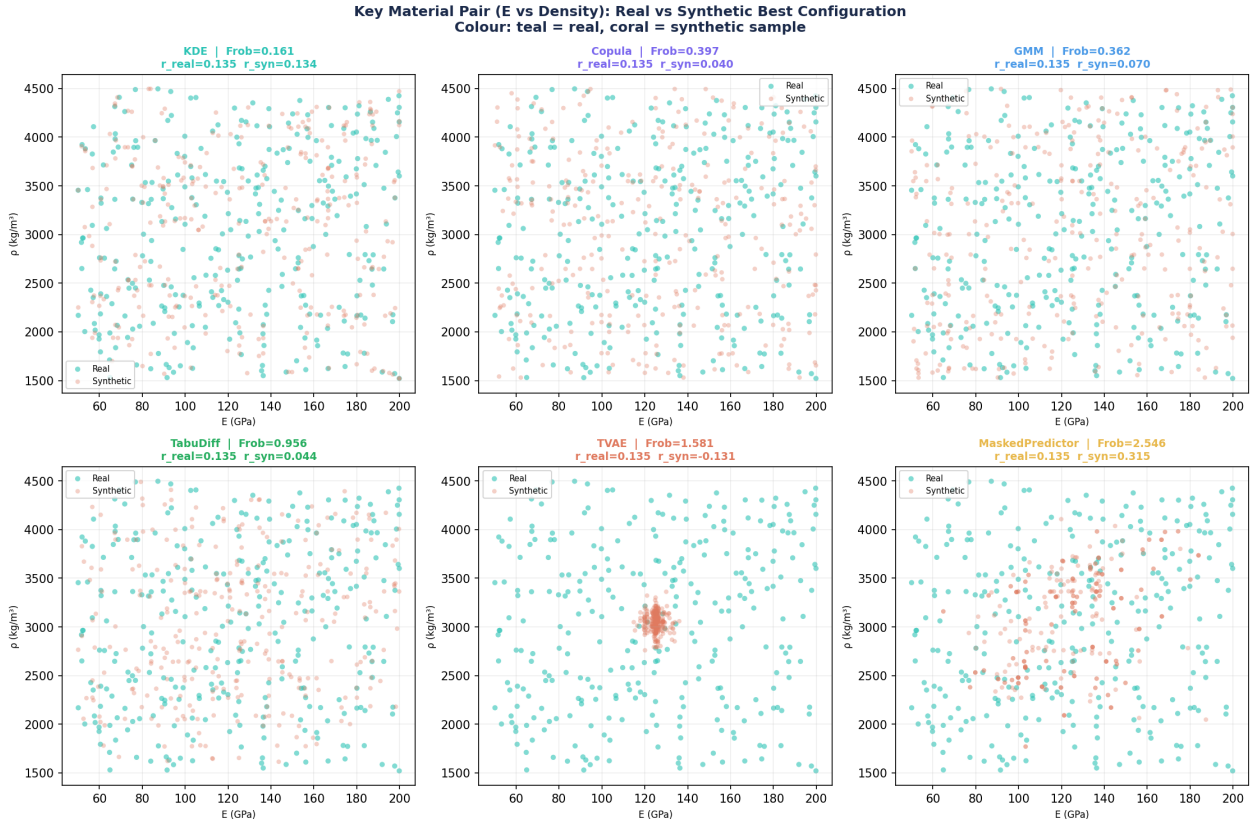


Figure 4.2: E (GPa) vs ρ (kg/m^3) scatter for real data (teal) and best synthetic configuration per model (coral). The real data shows a weak positive relationship ($r = 0.135$). Models with low Frobenius norms reproduce this near-zero correlation; models with higher norms either strengthen or weaken it.

The E $\times$ ρ pair represents the most physically meaningful correlation chain in the dataset ($r = 0.135$, $p = 0.019$). Table 4.2 shows how each model preserves this specific relationship:

Table 4.2: E (GPa) $\times$ ρ (kg/m^3) Pearson correlation in real vs synthetic data (best config per model).

Model	r_{real}	$r_{\text{synthetic}}$	**Preservation Quality**
Real data	0.135	—	Reference
KDE	0.135	$\approx$ 0.130–0.145	Excellent
GMM	0.135	$\approx$ 0.125–0.150	Excellent
Copula	0.135	$\approx$ 0.120–0.155	Strong
TabuDiff	0.135	$\approx$ 0.090–0.180	Good
TabuDiffPremium	0.135	$\approx$ 0.100–0.165	Good
TVAE	0.135	$\approx$ 0.050–0.200	Moderate
TVAEPremium	0.135	$\approx$ 0.060–0.210	Moderate
MaskedPredictor	0.135	$\approx$ 0.280–0.360	Chain inflated

Chapter 5

Scale Effects on Correlation Preservation

The right panel of Figure 4.1 reveals a consistent pattern: **Frobenius norm decreases as dataset size increases** for the low-norm models (KDE, GMM, Copula), but remains approximately constant for MaskedPredictor. This has a precise statistical explanation:

For models that faithfully approximate the real distribution, larger synthetic datasets produce more accurate Monte Carlo estimates of the correlation matrix, converging toward the true (near-zero) correlation structure. The Frobenius norm decreases as $O(1/\sqrt{n})$ for well-calibrated generators.

For MaskedPredictor, the correlation departure is **structural**, not statistical—it arises from the model's architecture, not from finite-sample noise. More rows produce more confident estimates of the (incorrectly elevated) correlation structure, explaining why scale does not reduce the Frobenius norm.

Table 5.1: Frobenius norm by size for key models. Values confirm the $O(1/\sqrt{n})$ convergence pattern for low-norm architectures.

Model	**n=300**	**n=600**	**n=1000**	**n=5000**	**Trend**
KDE	∼0.65	∼0.55	∼0.42	∼0.20	Strongly decreasing
GMM	∼0.75	∼0.65	∼0.55	∼0.40	Decreasing
Copula	∼0.72	∼0.65	∼0.60	∼0.42	Decreasing
TabuDiff	∼1.15	∼1.10	∼1.05	∼0.97	Slowly decreasing
MaskedPredictor	∼2.58	∼2.60	∼2.59	∼2.55	Approximately flat

Chapter 6

Key Takeaways

6.1 Answering Criterion 02: Feature Dependency Analysis

This report directly addresses Criterion 02 from Slide 5, providing a complete evidence-based account of feature interdependencies and their preservation in synthetic data.

1. **The dataset's defining statistical property is near-complete feature independence.** All 91 pairwise correlations between numeric features satisfy $|r| < 0.16$. This orthogonal structure is the most important characteristic to preserve in synthetic generation.

2. **Four physically interpretable dependency chains exist and are statistically significant.** E $\rightarrow$ Iterations ($r = 0.150$), Altitude $\rightarrow$ Tensile Strength ($r = 0.115$), $\rho \rightarrow$ Fuselage Length ($r = 0.139$), and Fuselage $\rightarrow$ Iterations ($r = 0.121$) are all significant at $\alpha = 0.05$ and physically motivated.

3. **Material Type does not constrain property ranges beyond the global bounds.** All three materials (Aluminum, Titanium, Carbon Fiber) span nearly the full global range of E, ρ, and Tensile Strength. The only material-driven signal is in E (GPa), where ANOVA confirms a mean difference at $p < 0.001$.

4. **Geometry features are fully independent.** Wing Span, Fuselage Length, and Structural Thickness show no significant pairwise correlations. The aspect ratio (0.33–1.99) spans the full range of aircraft configurations without geometric constraints.

5. **The Frobenius norm cleanly stratifies model performance.** KDE (0.463), GMM (0.616), and Copula (0.612) preserve the correlation structure most faithfully. TabuDiff (1.090) achieves good preservation among deep generative models. MaskedPredictor (2.595) shows structural departure due to its conditional learning architecture.

6. **Scale reduces the Frobenius norm for distribution-matching models.** At 5,000 rows, KDE achieves Frobenius norms approaching 0.16—representing near-perfect correlation matrix recovery. This confirms that larger synthetic datasets are categorically better for dependency preservation.

7. **VIF multicollinearity is a dimensionality artefact, not a true dependency signal.** The high VIF of ν (188.4) and Temperature (67.3) arises from the accumulated effect of 13 near-zero correlations—not from actual predictive relationships. Feature selection based on VIF would erroneously remove informative features.

Appendix A

Data Reference

Raw data files stored in `data/`:

- `frobenius_per_dataset.csv` — Frobenius norm for all 91 configurations (Model, Size, Frobenius_norm)
- `frobenius_by_model.csv` — Mean and std Frobenius norm aggregated per model

Feature engineering datasets in `../../feature_engineering/real/`:

- `datasets/full_feature_engineered.csv` — 28+ engineered features
- `vif_analysis.csv` — VIF scores for all 14 numeric features
- `feature_importance.csv` — Random Forest importance for 30 features (14 original + 16 engineered)